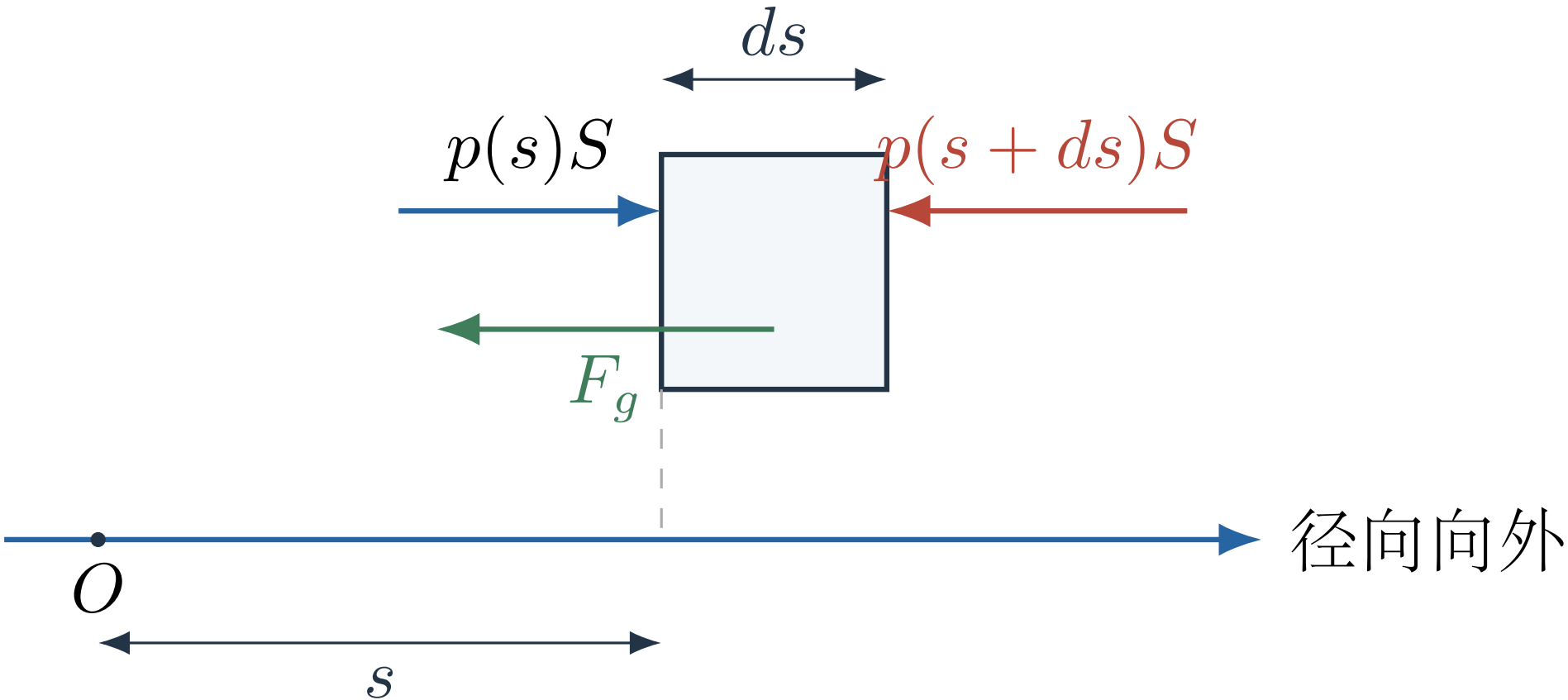

$$dm = \rho S\, ds, \quad F_g = dm\, g(s)$$